A BOUNDED ABOVE COMPLEX OF FLAT MODULES IS K-FLAT

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

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Editorial note. Both counted spans start at the \begin line of their environment, so the statement and the author's proof are delivered whole; surrounding prose of the section is deliberately outside the counted spans and is carried as standard prerequisites instead.

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Allowed standard prerequisites (editorial).

- K-flat complex, defined by the vanishing of the cohomology of the total tensor product with any acyclic complex. definition at source lines 14185-14191 of the same chapter. Supplies the property that is to be verified.
- The total tensor product of complexes is a direct sum in each degree, so a class in cohomology is represented by finitely many summands. used at proof lines 14290-14295 to replace L by a truncation. Reduces the general acyclic complex to a bounded above one.
- Spectral sequence of a first quadrant double complex and the resulting convergence statement. the author cites Homology, Lemma homology-lemma-first-quadrant-ss (tag 0132) at proof lines 14299-14302. Identifies the cohomology of the total complex with the limit of the spectral sequence whose first page vanishes.

Source reference. Original lemma 064K: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7869–7869. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 0132: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 1420–1420. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.2. *Let R be a ring. Let $P^\bullet$ be a bounded above complex of flat R-modules. Then $P^\bullet$ is K-flat.*

Proof. Let $L^\bullet$ be an acyclic complex of R-modules. Let $\xi \in H^n(\mathrm{Tot}(L^\bullet \otimes_R P^\bullet))$. We have to show that $\xi = 0$. Since $\mathrm{Tot}^n(L^\bullet \otimes_R P^\bullet)$ is a direct sum with terms $L^a \otimes_R P^b$ we see that ξ comes from an element in $H^n(\mathrm{Tot}(\tau_{\leq m} L^\bullet \otimes_R P^\bullet))$ for some $m \in \mathbf{Z}$. Since $\tau_{\leq m} L^\bullet$ is also acyclic we may replace $L^\bullet$ by $\tau_{\leq m} L^\bullet$. Hence we may assume that $L^\bullet$ is bounded above. In this case the spectral sequence of Homology, Lemma 0132 has

$$'E_1^{p,q} = H^p(L^\bullet \otimes_R P^q)$$

which is zero as P^q is flat and $L^\bullet$ acyclic. Hence $H^*(\mathrm{Tot}(L^\bullet \otimes_R P^\bullet)) = 0$. □